

MATHEMATICS Compulsory Part PAPER 2

11:30 am – 12:45 pm (1¼ hours)

INSTRUCTIONS

1. Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the ‘Time is up’ announcement.
2. When told to open this book, you should check that all the questions are there. Look for the words ‘**END OF PAPER**’ after the last question.
3. All questions carry equal marks.
4. **ANSWER ALL QUESTIONS.** You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.
5. You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.
6. No marks will be deducted for wrong answers.

There are 30 questions in Section A and 15 questions in Section B.
The diagrams in this paper are not necessarily drawn to scale.
Choose the best answer for each question.

Section A

1. $\frac{1}{k+2} + \frac{3}{5k-6} =$

A. $\frac{-8k}{(k+2)(5k-6)}.$

B. $\frac{-2k}{(k+2)(5k-6)}.$

C. $\frac{2k}{(k+2)(5k-6)}.$

D. $\frac{8k}{(k+2)(5k-6)}.$

2. $\frac{9^{3n+1}}{(3^{2n+3})(27^{2n+1})} =$

A. $9^{-n-2}.$

B. $9^{-n-1}.$

C. $9^{n-2}.$

D. $9^{n-1}.$

3. $(2\alpha - \beta)^2 + (\alpha - 2\beta)^2 =$

A. $3\alpha^2 - 4\alpha\beta + 3\beta^2.$

B. $3\alpha^2 - 8\alpha\beta + 3\beta^2.$

C. $5\alpha^2 - 4\alpha\beta + 5\beta^2.$

D. $5\alpha^2 - 8\alpha\beta + 5\beta^2.$

4. $230.045678 =$

- A. 230.0 (correct to 4 significant figures).
- B. 230.046 (correct to 4 significant figures).
- C. 230.0456 (correct to 4 decimal places).
- D. 230.04568 (correct to 4 decimal places).

5. If m and n are constants such that $(x-5)(x+m)-4(x+n) \equiv x(x+6)+1$, then $n =$

- A. -19 .
- B. -3 .
- C. 9 .
- D. 15 .

6. Solve $(x+2s)(x+t) = sx+st$, where s and t are constants.

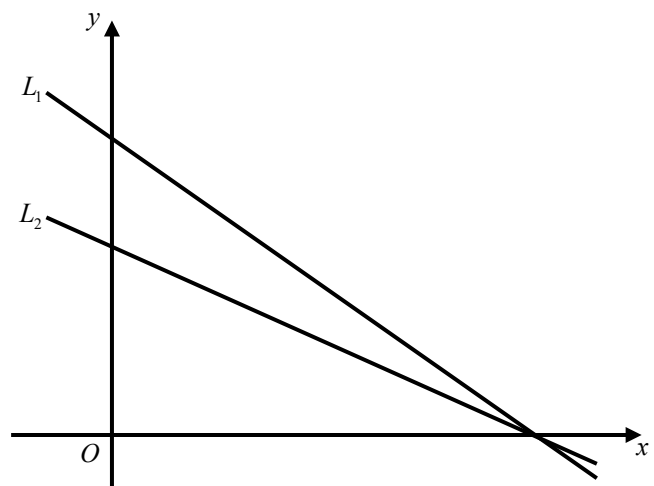
- A. $x = -s$
- B. $x = -2s$
- C. $x = -s$ or $x = -t$
- D. $x = -2s$ or $x = -t$

7. Let $f(x) = (x+d)(x-2)+9x$, where d is a constant. If $f(1+d) = 0$, then $d =$

- A. -2 .
- B. -1 .
- C. 1 .
- D. 2 .

8. Let $g(x) = hx^3 + 2x^2 + kx - 5$, where h and k are constants. If $g(x)$ is divisible by $x - 4$, find the remainder when $g(x)$ is divided by $x + 4$.
- A. -27
B. -10
C. 0
D. 54
9. The solution of $x - 1 > \frac{2x - 9}{3}$ or $3x + 12 \geq 0$ is
- A. $x \leq -4$.
B. $x \geq -4$.
C. $x < -6$.
D. $x > -6$.
10. The costs of rice of brand S and rice of brand T are \$12 / kg and \$18 / kg respectively. If s kg of rice of brand S and t kg of rice of brand T are mixed so that the cost of the mixture is \$16 / kg, then $s : t =$
- A. $1 : 2$.
B. $2 : 1$.
C. $2 : 3$.
D. $3 : 2$.
11. A sum of \$ P is deposited at an interest rate of 4% per annum for 5 years, compound quarterly. If the amount is \$80 000, find P correct to the nearest integer.
- A. 65 520
B. 65 564
C. 65 754
D. 66 667

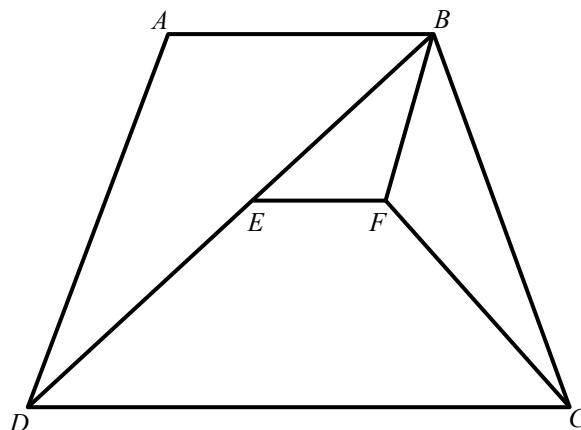
12. It is given that w varies directly as the square root of u and inversely as v . If u is increased by 44% and v is decreased by 20%, then w is increased by
- 40%.
 - 50%.
 - 64%.
 - 80%.
13. Let a_n be the n th term of a sequence. If $a_2 = 7$, $a_5 = 35$ and $a_{n+2} = 3a_{n+1} - 2a_n$ for any positive integer n , then $a_8 =$
- 67.
 - 131.
 - 259.
 - 515.
14. In the figure, the equation of the straight lines L_1 and L_2 are $ax + y = b$ and $cx + y = d$ respectively. It is given that L_1 and L_2 intersect at a point on the positive x -axis. Which of the following are true?
- $a < c$
 - $b > d$
 - $ad = bc$
- I and II only
 - I and III only
 - II and III only
 - I, II and III



15. Let p and q be real constants such that $pq < 0$. Which of the following statements about the graph of $y = (px+1)(qx+2)$ are true?
- I. The graph opens downwards.
 - II. The graph has two x -intercepts.
 - III. The graph passes through the origin.
- A. I and II only
 - B. I and III only
 - C. II and III only
 - D. I, II and III
16. The perimeter and the area of a sector are k cm and $3k$ cm² respectively. If the radius of the sector is 14 cm, then the arc length of the sector is
- A. 19 cm.
 - B. 21 cm.
 - C. 28 cm.
 - D. 49 cm.
17. The base of a right pyramid is a rectangle. The length and the breadth of the base are 18 cm and 16 cm respectively. The height of the pyramid is 12 cm. The pyramid is divided into a pyramid X and a frustum Y by a plane parallel to its base. If the breadth of the base of X is 12 cm, then the volume of Y is
- A. 486 cm³.
 - B. 666 cm³.
 - C. 688 cm³.
 - D. 944 cm³.

18. In the figure, $ABCD$ is a trapezium with $AB \parallel DC$ and $AB : CD = 3 : 5$. Let E be a point lying on BD such that $BE : ED = 2 : 3$. It is given that $AB \parallel EF$ and $AD \parallel BF$. If the area of $\triangle ABD$ is 225 cm^2 , then the area of the trapezium $CDEF$ is

- A. 222 cm^2 .
 B. 260 cm^2 .
 C. 279 cm^2 .
 D. 285 cm^2 .

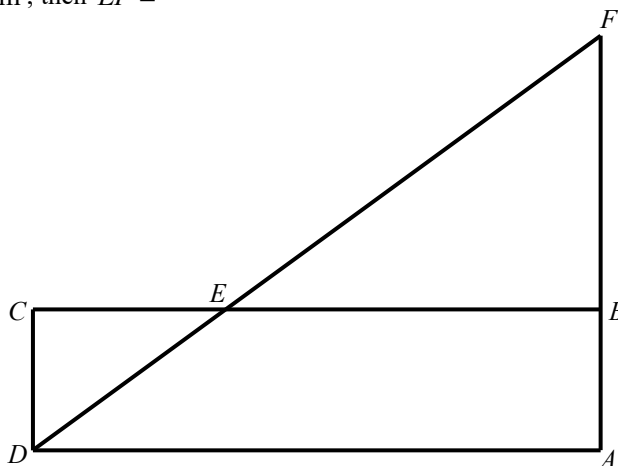


19. It is given that $\triangle PQR$ is a right-angled triangle with $\angle QPR = 90^\circ$. Denote the area of $\triangle PQR$ by S . If $QR = p$, $PR = q$ and $\angle PQR = \theta$, which of the following must be true?

- A. $S = \frac{1}{2} pq$
 B. $S = \frac{1}{2} pq \sin \theta$
 C. $S = \frac{1}{2} pq \cos \theta$
 D. $S = \frac{1}{2} pq \tan \theta$

20. In the figure, $ABCD$ is a rectangle. Let E be a point lying on BC . AB produced and DE produced meet at the point F . If $AB = 3 \text{ cm}$, $BE = 12 \text{ cm}$ and $CE = 4 \text{ cm}$, then $EF =$

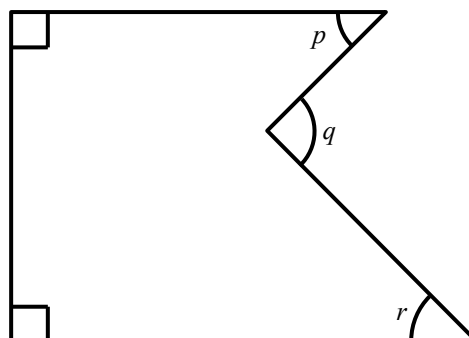
- A. 13 cm .
 B. 15 cm .
 C. 16 cm .
 D. 20 cm .



21. It is given that $WXYZ$ is a quadrilateral, where $\angle WXY = \angle YWZ$ and $\angle WYX = \angle WZY$. If $WX = 9$ cm, $WY = 40$ cm and $XY = 41$ cm, find the perimeter of the quadrilateral $WXYZ$.
- A. 100 cm
- B. 180 cm
- C. 410 cm
- D. 490 cm

22. According to the figure, which of the following must be true?

- A. $p = r$
- B. $q = 90^\circ$
- C. $p + r = q$
- D. $p + q + r = 180^\circ$



23. It is given that $ABCDEF$ is a regular 6-sided polygon. If AC and BF intersect at the point G , which of the following are true?

- I. $BG \parallel CE$
- II. $\triangle ABG \sim \triangle BDC$
- III. $\triangle AGF \cong \triangle BGC$

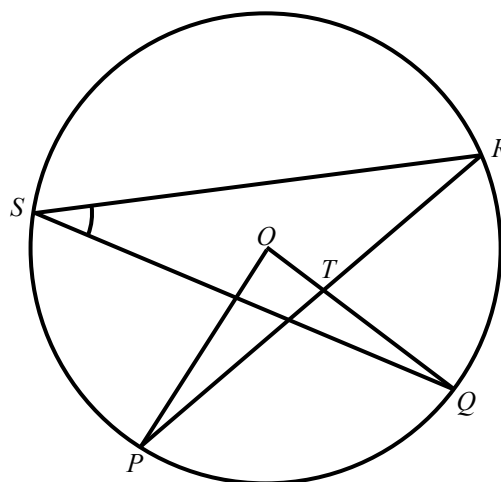
- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

24. The point $(3,1)$ is translated leftwards by 5 units to the point U . U is then rotated anticlockwise about the origin through 270° to the point V . Find the y -coordinate of V .

- A. -2
- B. -1
- C. 1
- D. 2

25. In the figure, O is the centre of the circle $PQRS$. PR and OQ intersect at the point T . If $\angle POT = 82^\circ$ and $\angle PTQ = 104^\circ$, then $\angle QSR =$

- A. 22° .
- B. 27° .
- C. 30° .
- D. 49° .



26. The equation of the straight lines L_1 and L_2 are $3x + 5y = 6$ and $6x - 10y = 13$ respectively. Let P be a moving point in the rectangular coordinate plane such that the perpendicular distance from P to L_1 is equal to the perpendicular distance from P to L_2 . The locus of P is

- A. a point.
- B. a circle.
- C. a square.
- D. a pair of straight lines.

27. The x -coordinate of the centre of the circle C is 5. If C passes through the points $(5,0)$ and $(11,4)$, then the circumference of C is

- A. 13π .
- B. 14π .
- C. 22π .
- D. 42π .

28. A bag contains five cards numbered 2, 2, 3, 4 and 6 respectively. Two cards are randomly drawn from the bag at the same time. Find the probability that the sum of the numbers drawn is greater than 7.

- A. $\frac{1}{2}$
B. $\frac{1}{5}$
C. $\frac{2}{5}$
D. $\frac{3}{10}$

29. Which of the following can be obtained from any box-and-whisker diagram?

- I. Mean
II. Lower quartile
III. Inter-quartile range

- A. I only
B. II only
C. I and III only
D. II and III only

30. Consider the following integers:

4 7 7 7 8 11 11 13 x y

Let u , v and w be the variance, the median and the mode of the above integers respectively. If both the mean and the range of the above integers are 9, which of the following must be true?

- I. $u < 7$
II. $v > 7$
III. $w = 7$

- A. I only
B. II only
C. I and III only
D. II and III only

Section B

31. $1010100000001110_2 =$

- A. $21 \times 2^{11} + 14.$
- B. $21 \times 2^{11} + 28.$
- C. $21 \times 2^{12} + 14.$
- D. $21 \times 2^{12} + 28.$

32. The H.C.F. and L.C.M. of three expressions are xy^2z^3 and $x^2y^4z^6$ respectively. If the first expression and the second expression are xy^3z^6 and xy^2z^4 respectively, then the third expression is

- A. $xy^2z^3.$
- B. $xy^4z^4.$
- C. $x^2y^2z^4.$
- D. $x^2y^4z^3.$

33. The sum of the first n terms of a sequence is $3^n - 1$. Which of the following are true?

- I. The 2nd term of the sequence is 8.
- II. The sequence is a geometric sequence.
- III. All the terms of the sequence are even numbers.

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

34. If β and $\frac{i^4 + 2i^3 + 3i^2 + 4i}{\beta i - 1}$ are real numbers, then $\beta =$

- A. -2 .
- B. -1 .
- C. 1 .
- D. 2 .

35. Which of the following statements about the graph of $y = 2 - \log_3(x + 4)$ is/are true?

- I. y increases when x increases
- II. The x -intercept of the graph is 5.
- III. The y -intercept of the graph is 2.

- A. I only
- B. II only
- C. I and III only
- D. II and III only

36. It is given that $\log y$ is a linear function of x . The intercepts on the vertical axis and on the horizontal axis of the graph of the linear function are 2 and 1 respectively. If $y = ab^x$, find a .

- A. 0.01
- B. 0.1
- C. 10
- D. 100

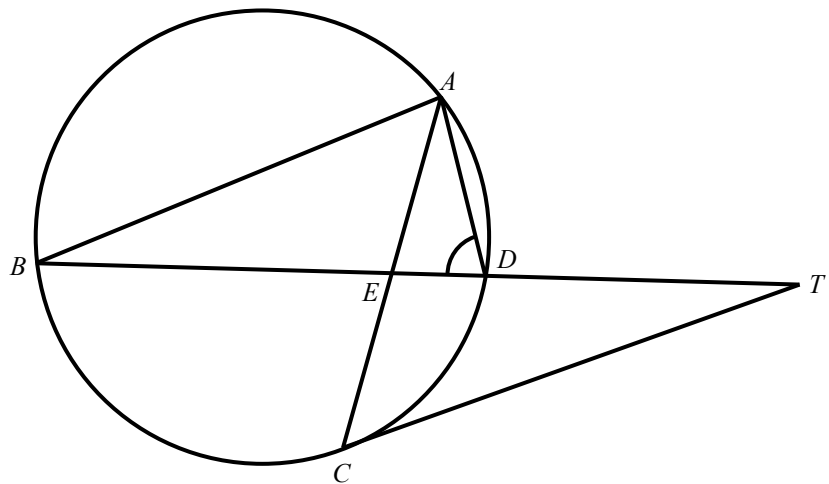
37. Consider the following system of inequalities:

$$\begin{cases} y \leq 7 \\ 3x - 4y \leq 2 \\ 6x + 13y \geq 25 \end{cases}$$

Let R be the region which represents the solution of the above system of inequalities. If (x, y) is a point lying in R , then the least value of $5x + 4y + 28$ is

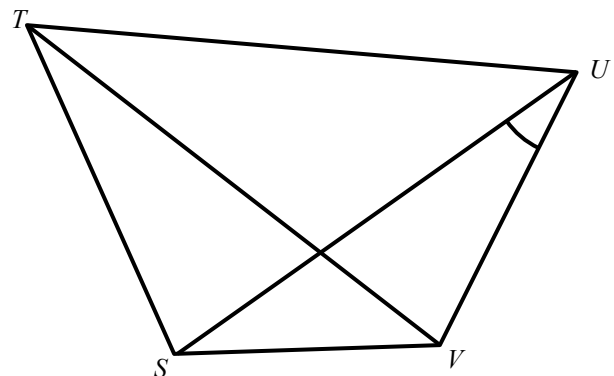
- A. 1.
B. 19.
C. 42.
D. 106.
38. In the figure, $ABCD$ is a circle. AC and BD intersect at the point E . BD is produced to the point T such that TC is tangent to the circle. It is given that $AB \parallel TC$. If $\angle AED = 85^\circ$ and $\angle CTE = 23^\circ$, then $\angle ADE =$

- A. 39° .
B. 56° .
C. 62° .
D. 72° .



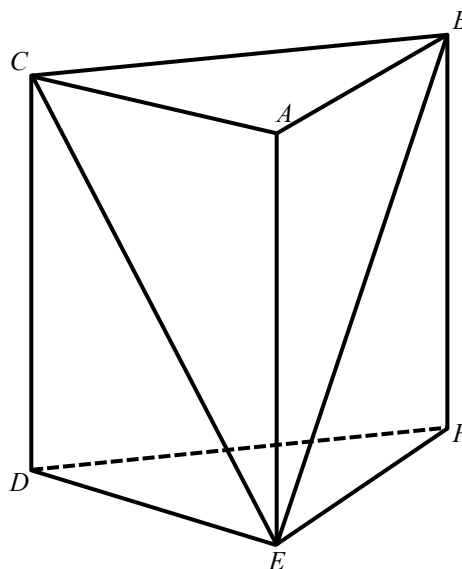
39. The figure shows the quadrilateral $STUV$, where $ST = 8$ cm and $SV = 5$ cm. If $\angle UTV = 30^\circ$ and $\angle TSU = \angle TVU = 90^\circ$, find $\angle SUV$ correct to the nearest degree.

- A. 22°
B. 28°
C. 30°
D. 38°



40. The figure shows the right prism $ABCDEF$, where $AB : AC : AE = 1 : 1 : 2$. Let θ be the angle between the plane ABC and the plane BCE . If $\angle BAC = 90^\circ$, find $\sin \theta$.

- A. $\frac{1}{3}$
 B. $\frac{2}{3}$
 C. $\frac{\sqrt{2}}{4}$
 D. $\frac{2\sqrt{2}}{3}$



41. For $90^\circ \leq \theta \leq 360^\circ$, how many roots does the equation $6\sin^4 \theta - 5\sin^2 \theta + 1 = 0$ have?

- A. 2
 B. 4
 C. 6
 D. 8

42. In a class, there are 3 boys and 7 girls. If 5 students are randomly selected from the class to form a committee, find the probability that at most 2 boys are selected.

- A. $\frac{5}{6}$
 B. $\frac{11}{12}$
 C. $\frac{119}{120}$
 D. $\frac{251}{252}$

43. There are 5 full-time employees and 3 part-time employees in a company. From the company, 7 employees are selected to form a queue. If no part-time employees are next to each other, how many different queues can be formed?

- A. 5 040
- B. 14 400
- C. 18 000
- D. 32 400

44. The table below shows the scores (in marks) and the corresponding standard scores of four students in a test.

Score (marks)	α	40	58	β
Standard score	-1	0	2	3

Find $\beta - \alpha$.

- A. 31
- B. 36
- C. 54
- D. 98

45. If the standard deviation of the five numbers p , q , r , s and t is 5, then the variance of the five numbers $\frac{p+2}{5}$, $\frac{q+2}{5}$, $\frac{r+2}{5}$, $\frac{s+2}{5}$ and $\frac{t+2}{5}$ is

- A. 1.
- B. 5.
- C. 9.
- D. 25.

END OF PAPER